

Kadir Cihan Duran - Weekly Reading Discussion

Generative Artificial Intelligence in Qualitative Data Analysis: Analyzing—Or Just Chatting? By Duc Cuong Nguyen and Catherine Welch

Duc Cuong Nguyen and Catherine Welch examine whether generative artificial intelligence (GenAI), specifically large language models (LLMs) like ChatGPT, can serve as legitimate scientific tools for qualitative data analysis. Software companies heavily promote these tools with promises that they can automate coding, identify qualitative themes, and speed up research projects by up to 30 times. Grounding their evaluation in the philosophy and sociology of science, the authors point out that scientific tools are never neutral; they actively shape the data we collect and define what counts as valid knowledge. Before a scholarly community accepts any new tool, it must prove that the tool is factually accurate, dependable, transparent, and ethically responsible. However, authors show that LLMs are merely autoregressive algorithms trained on vast internet text to predict the most statistically probable next word. Because they optimize for human-sounding plausibility rather than truth, LLMs are fundamentally indifferent to facts, highly unpredictable, and structurally opaque.

To test whether these models can genuinely analyze qualitative data, the authors conducted a hands-on evaluation using three sources of evidence: their own empirical testing, a cross-study comparison of five published papers, and an assessment of specialized qualitative data software. For their primary test, they uploaded a 49,444-word historical dataset into ChatGPT-4o and used various prompting strategies to generate codes and themes. Across all tests, it failed to perform real qualitative coding. Instead of indexing text line by line, ChatGPT produced broad, vague summaries, hallucinated quotations that never appeared in the uploaded text, and misattributed page numbers. When the authors tested dedicated qualitative software programs, such as NVivo and a startup tool called QUALSOFT, the tools fared no better: they produced banal themes, cut off participant quotes mid-sentence, conflated distinct individuals with companies, and concealed their algorithms behind black boxes.

The study reveals that instead of saving time, using GenAI traps researchers in an infinite loop of chatbot conversations, where every prompt produces a slightly different answer and forces the scholar to act as an unpaid content moderator verifying synthetic text. Synthesizing their findings, the authors identify five major epistemic risks: committing a category error by mistaking text generation for interpretation, producing unstable outputs, treating a chatbot as a human research partner, blaming user prompting rather than the tool's structural limits, and falling into the oracle effect by assuming machine outputs are objective. Crucially, the authors found an empirical paradox in the wider literature: even

when other researchers encountered severe hallucinations and dropped data, they rationalized these failures as creative alternative perspectives rather than recognizing them as tool malfunctions.

The paper concludes that qualitative data analysis is an intersubjective, reflexive, and deeply human craft centered on interpreting social meaning from real-world contexts. Autoregressive language models can easily mimic the visual appearance of qualitative research, such as neat coding tables and hierarchical data templates, but they possess zero understanding of human experiences or context. Nguyen and Welch warn that uncritically adopting these tools threatens to homogenize research, erode empirical rigor, and replace genuine interpretive analysis with superficial, automated chatting.

Discussion Questions

1. Can generative AI legitimately assist with early-stage brainstorming or literature synthesis without contaminating the reflexive, intersubjective nature of qualitative data analysis?
2. Does using LLMs actually accelerate qualitative workflows, or does the constant need to audit hallucinations and non-deterministic outputs simply replace interpretation with unpaid content moderation?
3. Why are researchers often inclined to rationalize obvious tool errors as "creative alternative perspectives" rather than holding AI systems to standard scientific criteria for accuracy and truth?